

HR Employee Attrition Analysis - Final Power BI Report

Excel | SQL | Power BI | Power Query | DAX | Star Schema

1. Project Overview

This project analyzes employee attrition using the IBM HR Employee Attrition dataset. The objective is to identify high-risk employee groups and present the findings through a clean, interactive Power BI dashboard.

Item	Details
Dataset Size	1,470 employee records
Main Objective	Identify attrition patterns by department, job role, overtime, age group, marital status, gender, and income
Dashboard Tool	Power BI Desktop
Modeling Approach	Star schema with one fact table and four dimension tables
Final Output	Interactive HR Attrition Analysis Dashboard

2. Tools and Techniques Used

Category	Tools / Techniques
Data Cleaning	Power Query, column removal, type checking, transformation steps
Data Modeling	Star schema, one-way relationships, fact and dimension tables
DAX Measures	COUNTROWS, CALCULATE, DIVIDE, AVERAGE
Visualization	KPI cards, bar charts, column charts, slicers
Previous Analysis	Excel Pivot Tables and SQL analysis in Google Colab / SQLite

3. Power BI Data Model

The original flat HR dataset was transformed into a simple star schema model. The raw source table was used as a reference source, while the cleaned employee-level table became the fact table.

Table	Purpose
Fact_Employee	Main employee-level fact table containing employee metrics, attrition status, age, income, and dimension keys
Dim_Department	Department lookup table
Dim_JobRole	Job role lookup table
Dim_Demographics	Gender, marital status, and age group lookup table
Dim_Work	Overtime, job satisfaction, and work-life balance lookup table

Relationship design: each dimension table filters Fact_Employee using one-to-many, single-direction relationships. This is a clean and interview-friendly modeling approach for the project.

4. Cleaning and Transformation Steps

- Imported only the raw HR employee dataset into Power BI.
- Created Fact_Employee from the raw data and dimension tables as references from the original dataset.
- Created ID/key columns for Department, Job Role, Demographics, and Work attributes.
- Merged dimension IDs back into Fact_Employee and removed duplicate text columns from the fact table.
- Kept Age in the fact table for numerical analysis and used Age Group in Dim_Demographics for grouping.
- Removed constant/non-analytical columns such as Over18 and StandardHours because they did not add analytical value.

5. Key DAX Measures

Measure	Logic / Purpose
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Total Employees	Counts all rows in Fact_Employee
Employees Left	Counts employees where Attrition = Yes
Active Employees	Counts employees where Attrition = No
Attrition Rate	Employees Left divided by Total Employees
Average Monthly Income	Average of MonthlyIncome
Average Age	Average of Age

6. Dashboard KPI Summary

KPI	Value
Total Employees	1,470
Employees Left	237
Active Employees	1,233
Overall Attrition Rate	16.12%

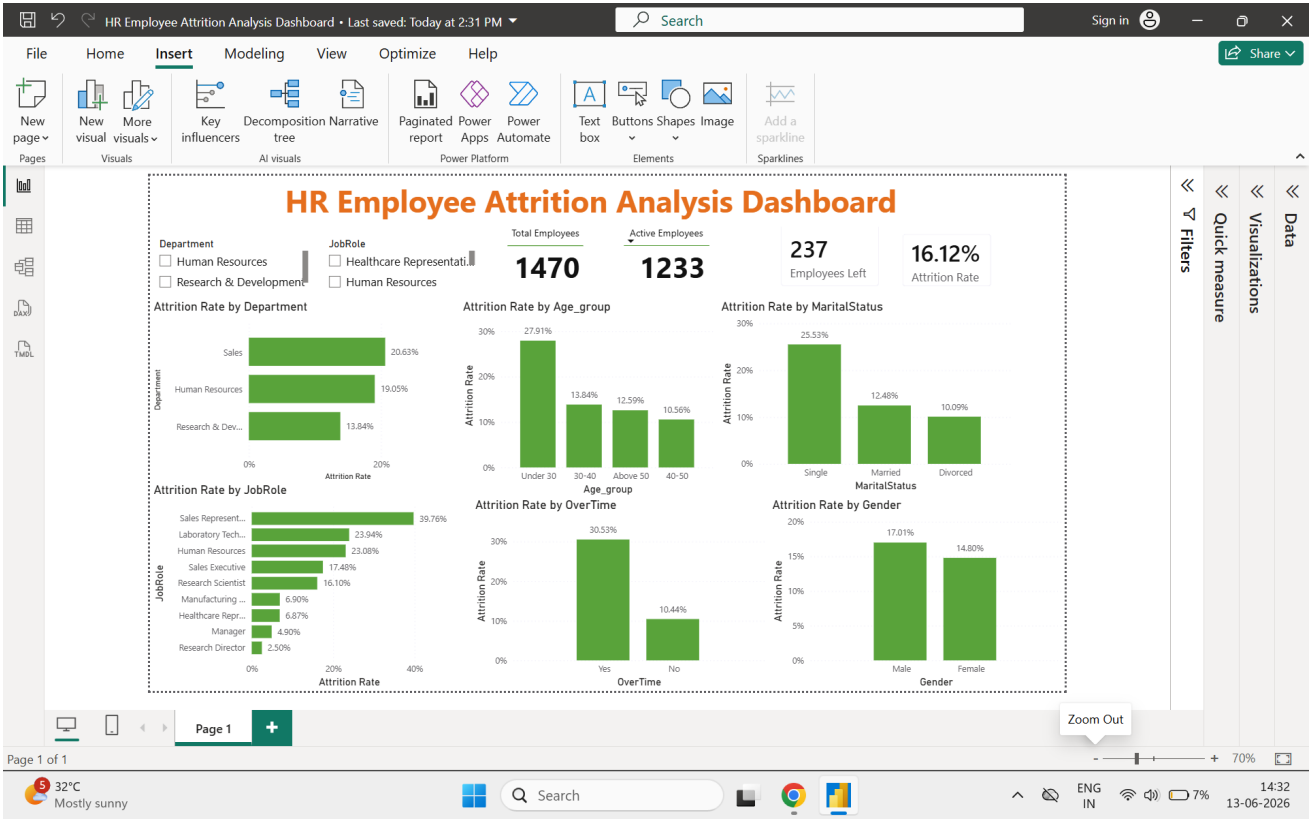
7. Dashboard Visuals Created

- Attrition Rate by Department
- Attrition Rate by Job Role
- Attrition Rate by Overtime
- Attrition Rate by Age Group
- Attrition Rate by Marital Status
- Attrition Rate by Gender
- Department slicer and Job Role slicer for interactivity

8. Key Business Insights

Area	Finding	Interpretation
Department	Sales has the highest attrition rate at 20.63%	Sales requires closer attention for retention planning
Job Role	Sales Representative has the highest attrition rate at 39.76%	This is the highest-risk job role in the dataset
Overtime	Employees doing overtime have 30.53% attrition compared with 10.44% for non-overtime employees	Overtime appears to be a strong attrition driver
Age Group	Under 30 employees show the highest attrition at 27.91%	Younger employees may require stronger engagement and growth opportunities
Marital Status	Single employees have the highest attrition at 25.53%	Single employees are a higher-risk demographic group
Gender	Male attrition is 17.01% and female attrition is 14.80%	Gender difference exists but is not the strongest attrition driver

9. Dashboard Preview



10. Recommendations

- Prioritize retention analysis for Sales Representatives and employees working overtime.
- Review workload distribution, overtime policies, and manager feedback in high-risk departments.
- Design engagement and career-growth initiatives for employees under 30.
- Monitor job roles with both high attrition and low average income, especially Sales Representative and Laboratory Technician.
- Use the Power BI dashboard as an interactive reporting tool for HR decision-making.

11. Final Project Status

The HR Employee Attrition project is complete and portfolio-ready. It includes Excel analysis, SQL analysis, a written report, and a Power BI dashboard with star schema modeling and DAX measures.

Checklist	Status
Excel dashboard and pivot analysis	Completed
SQL analysis in Google Colab / SQLite	Completed
Power BI star schema model	Completed
DAX KPI measures	Completed
Interactive dashboard with slicers	Completed
Final report	Completed